

Scalability & Future Plan

Date	04 July 2026
Team ID	SWTID-2026-9778
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited dataset size	May reduce prediction accuracy for diverse applicants	High
2	Local deployment only	Restricted accessibility and scalability	High
3	No real-time credit bureau integration	Limited validation of applicant credit information	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users	Deploy on cloud platform with load balancing to support thousands of users
2	Data Storage	Local SQLite database	Migrate to MySQL/PostgreSQL or cloud database for better scalability
3	Performance	Single machine model prediction	Optimize ML model, caching, and use GPU-enabled servers for faster predictions
4	Security	Basic user authentication	Implement JWT authentication, HTTPS, encryption, and role-based access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
-------	-------------------------------	-----------------	-----------------

Phase 1	Improve machine learning model accuracy using larger datasets and hyperparameter tuning	1-2 months	Higher prediction accuracy and reliability
Phase 2	Integrate real-time credit bureau and banking APIs	3-4 months	Better verification and more accurate approval decisions
Phase 3	Deploy the application on cloud platforms (AWS, Azure, or Google Cloud)	4-6 months	Improved scalability, availability, and accessibility
Phase 4	Develop a mobile application with advanced analytics dashboard	6-12 months	Better user experience and easier access for applicants and administrators